GEARED MOTOR SUN TRACKING SYSTEM FOR PARABOLIC TROUGH COLLECTOR

Final Report
by

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Abstract
This technical report details the development of a sun-tracking system for Parabolic Trough Collectors (PTC), an essential component in solar thermal energy systems. PTCs utilize parabolic mirrors to concentrate sunlight onto a receiver tube, where a heat transfer fluid absorbs the energy for electricity generation. Accurate sun tracking is vital for optimizing energy capture. The proposed system incorporates a single-axis tracking mechanism that follows the sun's movement throughout the day, ensuring maximum solar energy absorption. The design includes a motor and control algorithm for precise sun tracking, promoting efficient system performance. Additionally, the report covers torque calculations, chain and sprocket selection, motor selection, a SolidWorks model of the system and its components, material selection and comparisons, and the bills of materials.

Keywords: Parabolic Trough Collector (PTC), Sun Tracking System, Single-Axis Mechanism, Solidworks, chain and sprocket

Section 1: Introduction

Solar energy stands as one of the most abundant and sustainable sources of power available today, with Parabolic Trough Collectors (PTC) being a widely used technology in solar thermal energy systems due to their efficiency in harnessing solar radiation. PTCs employ parabolic mirrors to focus sunlight onto a receiver tube, where a heat transfer fluid[1] absorbs the energy for electricity generation and various industrial applications. The performance of PTCs heavily relies on accurate sun tracking to optimize solar energy absorption[2]. This report presents the development of a sun-tracking system for PTCs aimed at improving energy efficiency through precise alignment with the sun throughout the day. The report details the fundamental principles of PTC operation, explores the significance of tracking systems, examines tracking algorithms and associated challenges, and outlines the design methodology for implementing an effective sun-tracking mechanism.

In this report the Section 2 goes through the importance of the topic, Section 3 explores details of the PTC, section 4 is about the design of tracking mechanism, section 5 speaks about the Bill of Materials, section 6 is about other factors like safety, durability and operation, and then we conclude the report work.

Section 2: Topic and its Importance

Our topic is the sun tracking system for Parabolic Trough Collectors (PTC), which plays a vital role in enhancing the efficiency of solar thermal energy systems. PTCs use curved, reflective surfaces to focus sunlight onto a receiver tube, where the concentrated solar energy heats a working fluid to generate electricity or provide thermal energy for industrial applications [2]. The sun tracking system ensures that the collectors remain optimally aligned with the sun throughout the day, maximizing the capture of direct beam radiation.

Importance of the Sun Tracking System [3]:
1. Increased Energy Efficiency: By dynamically adjusting the collector’s orientation to follow the sun’s path, tracking systems significantly improve energy absorption compared to fixed solar collectors.
2. Optimal Solar Utilization: Sun tracking systems maximize the utilization of available solar energy, particularly in regions with high levels of direct normal irradiance (DNI).

3. Cost-effective Energy Production: Improved efficiency reduces the cost per unit of energy produced, making solar power more economically viable for various applications.
4. Sustainable Energy Solutions: By increasing the solar energy capture, these systems contribute to the development of sustainable and environmentally friendly energy infrastructure.[3][4]
5. Technological Advancements: Modern tracking systems incorporate control algorithms and automated mechanisms to precisely track the sun, enhancing reliability and performance.[3]

Subsection 2.1: Classification based on Movement Type

Solar tracking systems are classified as passive or active based on their movement mechanism. Passive trackers use thermal expansion to follow the sun without motors or controllers, making them simple and low-cost but less accurate—suitable mainly for PV systems. Active trackers, ideal for PTCs, use motors and controllers (with sensors or algorithms) to provide precise, real-time sun alignment. While more accurate and efficient, active systems are also more complex and costly. [5].

Subsection 2.2: Classification based on Control Strategy

Solar tracking systems can also be categorized by their control strategies, which determine how the system identifies the sun's position and adjusts the collector's orientation. These strategies greatly influence system accuracy, complexity, and cost, and are typically classified into three types:

a) Open-loop Control: Open-loop systems use predefined algorithms—such as the Solar Position Algorithm (SPA)—to calculate the sun's position based on time, date, and geographic location, without real-time feedback. They are simple, cost-effective, and weather-independent, but may lose accuracy due to external factors like dirt, wind, or mechanical shifts. This control method is adopted in our design. **This will be the type of control that we will be using in our design.**

b) Closed-Loop Control: Closed-loop systems rely on real-time feedback from sensors (e.g., LDRs, GPS, photodiodes) to dynamically adjust the collector's orientation. They offer high accuracy and adaptability but are more complex and costly, with potential performance issues under cloudy conditions. [3][4][5]

c) Hybrid-Loop Control: Hybrid systems integrate both open-loop algorithms and sensor feedback, combining predictive accuracy with real-time correction. While they enhance efficiency and reliability in variable conditions, they also introduce greater system complexity and cost..

Using a gravity battery in place of an active open-loop control system for tracking in a Parabolic Trough Collector (PTC) has a few advantages but significant limitations[6][7]. The potential benefits are that it does not require a continuous energy supply during operation and does not require batteries to store energy.
But the limitations of it are:

- Precision and Responsiveness: Active open-loop control (with motors + programmed algorithms) can precisely follow the solar angle through the day, whereas gravity systems are limited to preset movements, making them less accurate in aligning with the sun throughout the year.
- Limited Control Modes: Gravity systems can't easily implement emergency stow positions (during storms), which is a standard feature in modern PTC control logic.

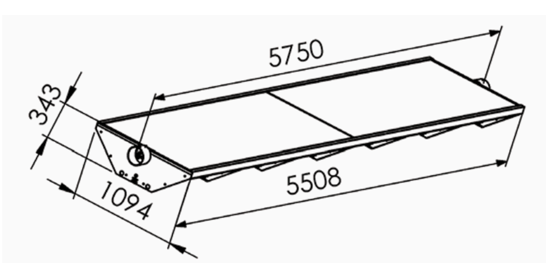
- Complex Mechanical Design: Although electrically simpler, building a gravity-based system that can track large, heavy collectors like PTCs with adequate torque and stability is mechanically challenging. Also, gravity storage requires vertical shafts or towers, which may not be feasible near all PTC installations.
- Reset Complexity: Active systems automatically reset at dawn. Gravity batteries would need manual or auxiliary power to lift weights overnight, adding operational complexity.

Section 3: Details of Parabolic Trough Collector Selected

The tracking system that we will design will be designed for Absolicon T160 PTC. The details for the selected PTC are mentioned below:

Table 1: Specifications of PTC [8]

Model Name	Absolicon T160
Collector Type	Glass-covered parabolic trough collector with one-axis tracking
Recommended Heat Transfer Fluid	Water, Propylene Glycol (maximum 40%)
Volume of Heat Transfer Fluid	2.2 litre
Operational Temperature	60-160°C
Stagnation Temperature	460°C
Maximum Operating Pressure	16 bar
Receiver	Stainless steel, optically selective coating
Glass	4 mm hardened glass, anti-reflective coating
Reflector	Polymer-embedded silver on steel sheet
Weight	148 kg
Dimensions	5508 mm X 1094 mm X 343 mm



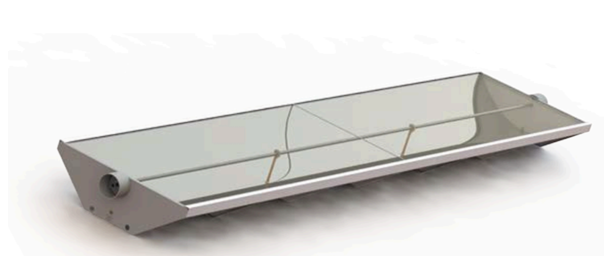


Figure 2: Selected PTC [8]

Section 4: Design of tracking mechanism

This section includes different components and calculations that were done to design a tracking mechanism for the selected PTC.

Subsection 4.1: Torque Calculation

<u>Torque calculation due to the movement of PTC and support structure</u>

Given above is our selected PTC; the equation of the curvature of it is as follows:

$$x^2 = 4 \cdot 218.0823 \cdot y \text{ ; where x \& y are in mm} \tag{1}$$

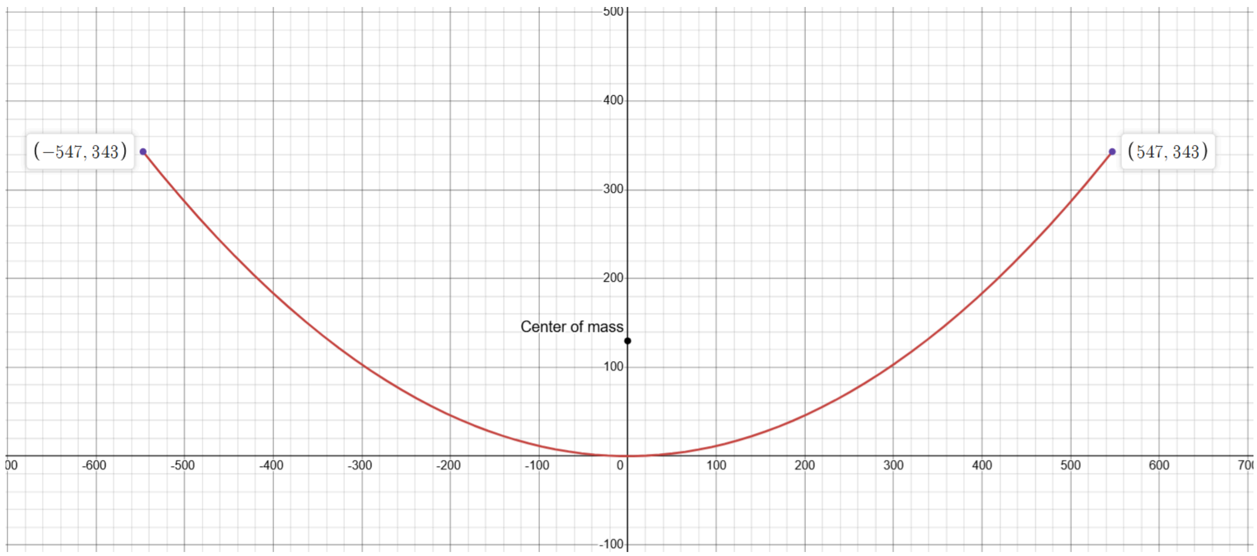


Figure 3: Equation parabola plotted on Desmos[9]

The center of mass is calculated by the centroid formula.

$$y_{com} = \frac{\int_{-547}^{547} \left(\frac{x^2}{4\cdot 218.0823}\right)\sqrt{1+\left(\frac{x}{4\cdot 218.0823}\right)^2}\,dx}{\int_{-547}^{547} \sqrt{1+\left(\frac{x}{4\cdot 218.0823}\right)^2}\,dx} \tag{2}$$

Thus, we get $y_{com} = 129.796\ mm$.

We calculated the moment of inertia assuming uniform mass density along the curve shown in the above figure, given that the structure weighs 148 kg. We used the method of Mass Moment of Inertia of a thin wire or curve to calculate the moment of inertia.

The mass density is calculated as,

$$\mu_{PTC} = \frac{148}{\int_{-547}^{547} \sqrt{1+\left(\frac{x}{4\cdot 218.0823}\right)^2}\,dx} \tag{3}$$

$$\mu_{PTC} = 0.110825\ kg/mm = 110.8\ kg/m \tag{4}$$

The moment of inertia is calculated using the perpendicular axis theorem and the parallel axis theorem. We calculate the I_x and I_y are the moments of inertia about the x-axis and y-axis as shown in fig. 3.

$$I_x = \int_{-547}^{547} \left(\frac{x^2}{4\cdot 218.0823}\right)^2 \sqrt{1 + \left(\frac{x}{4\cdot 218.0823}\right)^2}\, dx \quad (5)$$

$$I_y = \int_{-547}^{547} x^2 \sqrt{1 + \left(\frac{x}{4\cdot 218.0823}\right)^2}\, dx \quad (6)$$

$$I_z = I_x + I_y \quad (7)$$

And using parallel axis theorem to make the axis of rotation as the axis of center of mass (parallel to z-axis),

$$I_{com,z} = I_z - m \cdot (y_{com})^2 \quad (8)$$

Thus calculated $I_{com,z}$ comes to be, $I_{com,z} = 18.407\ kg\ m^2$

As mentioned in the section below, our selected motor is a high-torque stepper motor, which possesses a movement profile as below for every step.

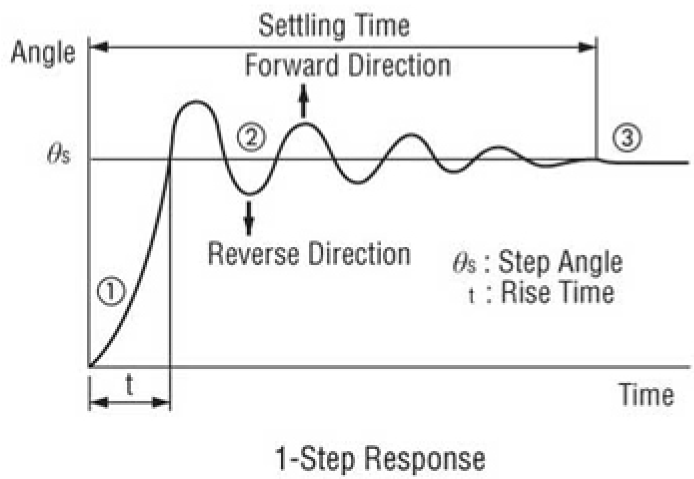


Figure 4: 1-step Response of a stepper motor [10]

A typical rise time of a stepper motor with high torque is ~22 ms (considering the values of phase resistance (R): 0.8 Ω and phase inductance (L): 8 mH = 0.008 H). Thus, the calculated movement torque will be. (assuming the maximum velocity achieved is 2*average angular velocity, i.e., 0.2 deg/sec)

$$max\ torque = I\left(\frac{dw}{dt}\right)_{max} \quad (9)$$

$$\left(\frac{dw}{dt}\right)_{max} = \frac{0.2}{22\times 10^{-3}} \frac{\pi}{180} = 0.1586\ rad/s^2$$

$$max\ torque\ due\ to\ movement = 18.407 \times 0.1586$$

$$max\ torque\ due\ to\ movement = 2.9193\ Nm$$

<u>Torque calculation due to wind</u>

A major amount of torque will be needed to account for the torque due to wind. We know the net flow-induced moment J is [11],

$$J_{wind} = 0.5 \cdot C_{moment} \cdot \rho \cdot v^2 \cdot A_{ref} \cdot l_{ref} \quad (10)$$

Where,

C_{moment} = Moment coefficient, i.e., constant for particular shape and flow conditions = 0.2
ρ = Air density = 1.229 kg/m^3
v = Wind velocity = 10 m/s
A_{ref} = Reference area of the collector (aperture area) = 6.025 m^2
l_{ref} = Reference length (collector aperture width) = 1.094 m
The torque required to overcome the wind is,

$$J_{wind} = 81 \text{ Nm}$$

Subsection 4.2: Support structure
The support structure of a PTC system with single-axis solar tracking is designed for high mechanical stability, optical precision, and environmental durability. The structural and tracking components are constructed from materials selected for strength, corrosion resistance, and thermal compatibility.

1. Absorber Tube Mounting System: The receiver tube, typically a stainless steel tube with a selective coating, is supported at the focus using 304 or 316 stainless steel U-bolts or V-saddles mounted on aluminium or galvanized steel brackets. Expansion joints made from Inconel or flexible stainless steel allow for thermal expansion due to temperature variations. The support design minimizes shading and reduces thermal bridging losses.
2. Rotational Axle and Bearings: Single-axis tracking is accomplished by rotating the trough about a high-strength carbon steel shaft (commonly AISI 1045), which is supported by ball bearings with cast iron housings (e.g., FNL series bearings by SKF). These bearings are designed to accommodate axial and radial loads and allow smooth and precise movement of the structure. The outer diameter of the ball bearings used is equal to the hub diameter of the bigger sprocket, i.e., 3.75inches. The rotational axle used is of the same diameter as the hub diameter of the smaller sprocket, i.e., 1.25inches.
3. Support Piers and Foundations: The tracking shaft and trough structure are supported by reinforced concrete piers made using M30–M35 grade concrete or galvanized steel columns (e.g., ASTM A500 Grade B) embedded into a concrete foundation. The foundation design follows local wind and seismic load standards (e.g., ASCE 7-16) to ensure structural safety.
4. Drive and Tracking Mechanism: The tracking system typically consists of a stepper motor (e.g., NEMA 34 MST513) or DC linear actuator connected to a worm gear slewing drive (e.g., IMO WD series or IGUS slewing ring gears). The actuator is controlled via a microcontroller or PLC using astronomical solar tracking algorithms or feedback from sun position sensors (e.g., Kipp & Zonen CHP1).
5. Bracing and Structural Reinforcements: Wind and torsional loads are resisted using galvanized steel diagonal bracings (e.g., L-profiles or tubular members made from ASTM A500 steel) and torsion-resistant trusses. In large installations, dampers may be used to suppress oscillations caused by wind gusts.
6. Cabling and Controls: Electrical and control wiring is enclosed in UV-stabilized PVC conduits, corrugated flexible conduits, or galvanized cable trays. Cables are rated for outdoor temperature extremes and UV exposure (e.g., XLPE-insulated cables).

Subsection 4.3: Motor selection and gear ratio calculation
<u>Motor selection</u> [13]

Since, our application requires a high torque but very low-speed motor. The motor that we have selected to rotate our gears (and ultimately, the PTC) precisely is a stepper motor MST513, which has a minimum step size of 1.8° with a step accuracy of ± 5%. To reduce the step error and gear ratio, we will rotate the rotor of the stepper motor by an angle of 90º. So, the shaft of the rotor will rotate by 90º in 5 seconds, and after going through the gear transmission system, the PTC will rotate by 0.5º in 5 seconds.

Figure 5: Motor MST513 [13]

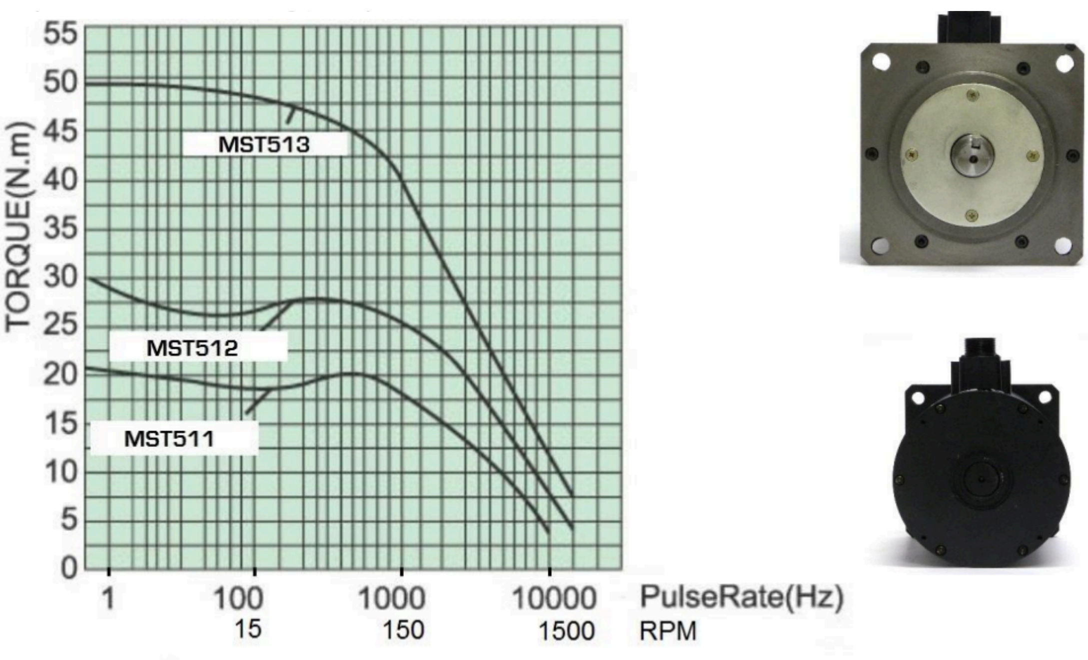


Figure 6: Torque speed curve of selected motor [13]

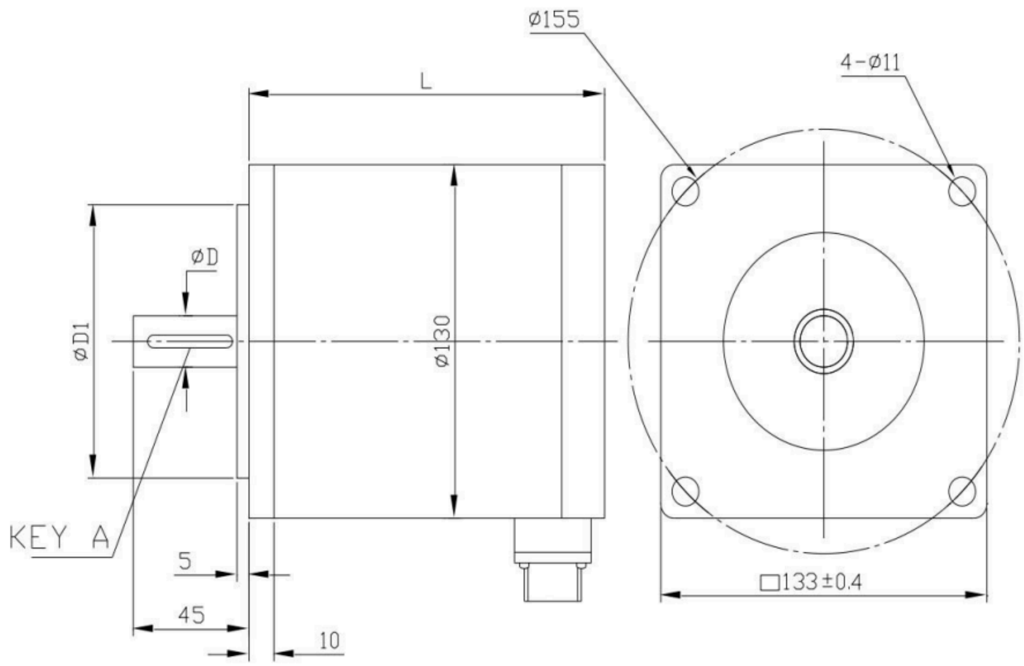


Figure 7: Engineering Drawing of the selected motor [13]

Model no	No. of phase	Step Angle	Rated Voltage	Current / Phase	No Load Run Frequency	Holding Torque	Rotor Inertia	Length	Shaft Dimension Φ D		Flange Dimension Φ D		KEY A	Wiring Diagram
Single shaft		degree	V	A	KHz	N.m	Kg-cm²	mm	mm		mm			
MST511C213 -X2AA6.0	2	1.8	120-310	6.0	≥20	27	33	165	19	$^{-0.013}_{-0.028}$	100	$^{0}_{-0.023}$	5x25	1
MST512C213 -X2AA7.0	2	1.8	120-310	7.0	≥15	40	48	230	19	$^{-0.013}_{-0.028}$	100	$^{0}_{-0.023}$	5x25	1
MST513C213 -X2AA7.0	2	1.8	120-310	7.0	≥12	50	60	270			100	$^{0}_{-0.023}$		1
On request	3	1.2	80-325	6.0	≥15	25	33	165	19	$^{-0.013}_{-0.028}$	100	$^{0}_{-0.023}$	5x25	
On request	3	1.2	80-325	6.0	≥15	37	48	230	19	$^{-0.013}_{-0.028}$	100	$^{0}_{-0.023}$	5x25	
On request	3	1.2	80-325	6.0	≥15	50	60	270			100	$^{0}_{-0.023}$		
On request	5	0.72	120-310	5.0	≥20	20	33	165	19	$^{-0.013}_{-0.028}$	100	$^{0}_{-0.023}$	5x25	2
On request	5	0.72	120-310	5.0	≥20	30	48	230	19	$^{-0.013}_{-0.028}$	100	$^{0}_{-0.023}$	5x25	2
On request	5	0.72	120-310	5.0	≥15	40	60	270			100	$^{0}_{-0.023}$		2

Figure 8: Specification of selected motor [13]

<u>Gear ratio calculation</u>

To calculate gear ratio, we fix the number of steps that need to be rotated for 5 seconds which in this case is 50 steps, therefore the shaft of the motor will rotate by 90° which should result in a 0.5° rotation of PTC after gear system transmission.

$$\therefore \frac{90}{0.5} = 180$$

Therefore, we will require a 1:180 gear ratio, which will be achieved by attaching one planetary gearbox of gear ratio 1:20 at the output of the shaft of the motor. This will be attached to the 1:3 chain and sprocket mechanism, which will again be attached to the 1:3 chain and sprocket mechanism. Hence fulfilling our torque and RPM requirements for PTC.

Subsection 4.4: Chain and sprocket design

The chain and sprocket system were chosen according to the torque and load requirements. The required torque for the 83.9 Nm, thus we choose the chain and sprocket system according to the American National Standards Institute (ANSI) Standards. [14]

We have used No. 50 (5/8" pitch) [15] chains and sprockets, which are used for heavy load applications because of their robust mechanical properties and torque-handling capability.

Table 2: Specification of the Chain Sprocket System[15]

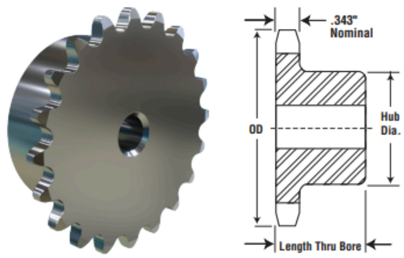
Feature	Benefit
#50 chain	Strong, reliable, well-supported size for industrial loads
⅝" or 0.625" (inch) pitch	Ensures tight, stable meshing and reduced slip
60T sprocket (T = tooth)	Distributes load, increases torque, reduces wear
20T driver	Allows speed reduction and torque multiplication
3:1 ratio	Ideal balance for power transmission efficiency and durability

We have chosen 60T and 20T sprockets because we need a ratio of 1:3 to increase the torque and reduce the speed on the Parabolic Trough Collector.

Using the above information, we now proceed to choose the sprocket type. To do so, we have used the Roller Chain Sprockets Catalog (Pg 46). [15]

No. 50 **5/8" Pitch**

Stainless Steel Stock Sprockets



Stainless Steel — Type B

Alteration Charges

See current discount sheet for alteration charges.

Single - Type B — Stainless / **Single - Type A**

No. Teeth	Catalog Number	OD	Type	Bore		Hub		Wt. lb (Approx.)	Type	Catalog Number	Stock Bore	Wt. lb (Approx.)
				Stock	Rec. Max	Dia.	LTB					
8	50B8SS	1.884	B	.625	.625	1.125	1	0.25				
9	50B9SS	2.093	B	.625	.75	1.375	1	0.36				
10	50B10SS	2.300	B	.625	.875	1.563 ★	1	0.50				
11	50B11SS	2.500	B	.625	1	1.75 ★	1	0.60				
12	50B12SS	2.710	B	.625	1.25	1.984 ★	1	0.70				
13	50B13SS	2.910	B	.625	1.313	1.875	1	0.80	A	50A13SS	.625	0.42
14	50B14SS	3.110	B	.625	1.438	2.125	1	1.00	A	50A14SS	.625	0.50
15	50B15SS	3.320	B	.625	1.5	2.375	1	1.30	A	50A15SS	.625	0.54
16	50B16SS	3.520	B	.625	1.75	2.5	1	1.50	A	50A16SS	.625	0.68
17	50B17SS	3.720	B	.625	1.875	2.688	1	1.80	A	50A17SS	.625	0.76
18	50B18SS	3.920	B	.625	1.875	2.875	1	2.00	A	50A18SS	.625	0.86
19	50B19SS	4.120	B	.625	1.75	2.5	1	2.23	A	50A19SS	.625	0.94
20	50B20SS	4.320	B	.75	1.75	3	1	2.30	A	50A20SS	.75	1.06
21	50B21SS	4.520	B	.75	2	3	1	2.42	A	50A21SS	.719	1.40
22	50B22SS	4.720	B	.75	2	3	1	2.54	A	50A22SS	.719	1.60
23	50B23SS	4.920	B	.75	2	3	1	2.67	A	50A23SS	.719	1.70
24	50B24SS	5.120	B	.75	2	3	1.25	3.38	A	50A24SS	.719	1.80
25	50B25SS	5.320	B	.75	2	3	1.25	3.42	A	50A25SS	.719	1.90
26	50B26SS	5.520	B	.75	2	3	1.25	3.57	A	50A26SS	.719	1.70
28	50B28SS	5.920	B	.75	2	3	1.25	3.88	A	50A28SS	.719	2.50
30	50B30SS	6.320	B	.75	2.25	3.25	1.25	4.54	A	50A30SS	.719	2.70
32	50B32SS	6.721	B	.75	2.25	3.25	1.25	4.96	A	50A32SS	.719	2.72
35	50B35SS	7.320	B	.75	2.25	3.25	1.25	5.44	A	50A35SS	.719	3.70
36	50B36SS	7.519	B	.75	2.25	3.25	1.25	5.64	A	50A36SS	.719	3.82
40	50B40SS	8.320	B	.75	2.25	3.25	1.25	6.50	A	50A40SS	.719	4.70
45	50B45SS	9.310	B	.75	2.5	3.75	1.25	8.50	A	50A45SS	.719	6.00
48	50B48SS	9.911	B	1	2.5	3.75	1.25	9.28	A	50A46SS	.938	6.58
54	50B54SS	11.106	B	1	2.5	3.75	1.25	11.00	A	50A54SS	.938	8.30
60	50B60SS	12.300	B	1	2.5	3.75	1.25	14.00	A	50A60SS	.938	10.80

Figure 9: Pg 46 of Chain Sprocket Catalog [15]

The above figure shows the section for Chain #50 and pitch 0.625 inches. Now, since we know the ratio, we choose 60T and 20T, and accordingly, we chose the outer diameter, bore, and hub size.

Table 3: Sprocket specifications [15]

Parameter	60T Sprocket (50B60SS)	20T Sprocket (50B20SS)
Type	B	B
Outside Diameter (OD)	12.300 in	4.920 in
Stock Bore	1.00 in	0.75 in
Recommended Max Bore	2.5 in	1.75 in

Hub Diameter	3.75 in	2.00 in
Length Thru Bore (LTB)	1.25 in	1.25 in
Weight (Approx.)	10.80 lb	2.54 lb

ANSI sprockets are notably characterized by 'type', which indicates the sprocket's hub style. Hub styles offer unique mounting shaft mounting configurations, and there are four styles [14]:

1. Style A - a flat sprocket with no hub
2. Style B - a sprocket with a hub on one side
3. Style C - a sprocket with hubs on both sides
4. Style D - a sprocket with a bolted hub attached to a plate

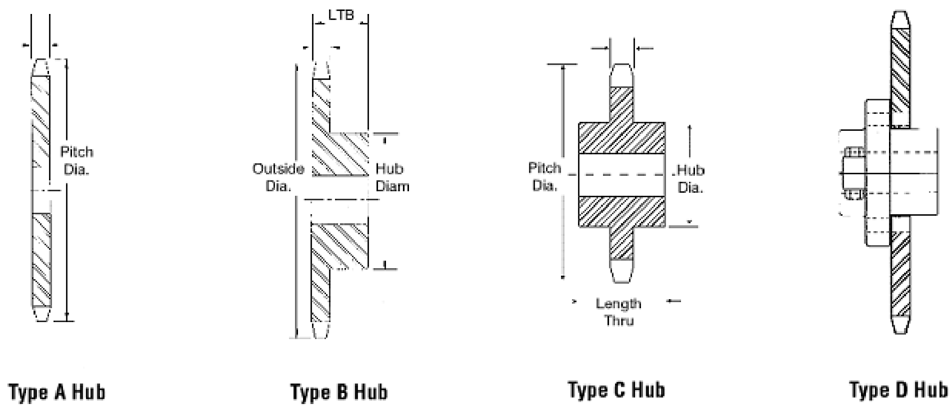


Figure 10: Types of Hub according to ANSI Standards [14]

For our purpose, we are choosing hub type B. Hub-type B sprockets are commonly preferred for heavy loads and industrial applications[14] due to their design features and mechanical advantages, particularly in terms of strength, alignment, and durability. [16]

Table 4: Features of type B Hub [15][16]

Feature	Benefit
Thicker hub on one side	Provides extra material for a stronger keyway and shaft connection.
Improved load distribution	Helps in better torque transmission without shaft deformation.

Wider face contact area	Ensures greater clamping strength and less chance of slippage.
Easier shaft mounting	A hub on one side allows easy access for installation and maintenance.
Accommodates keyways/set screws	More space to securely fit keyways, bushings, or set screws.
Better alignment support	Helps maintain parallel alignment, reducing chain wear and vibration.

Using the above information and sizing, have designed (Fig 11) a model of the chain sprocket system on Solid Works

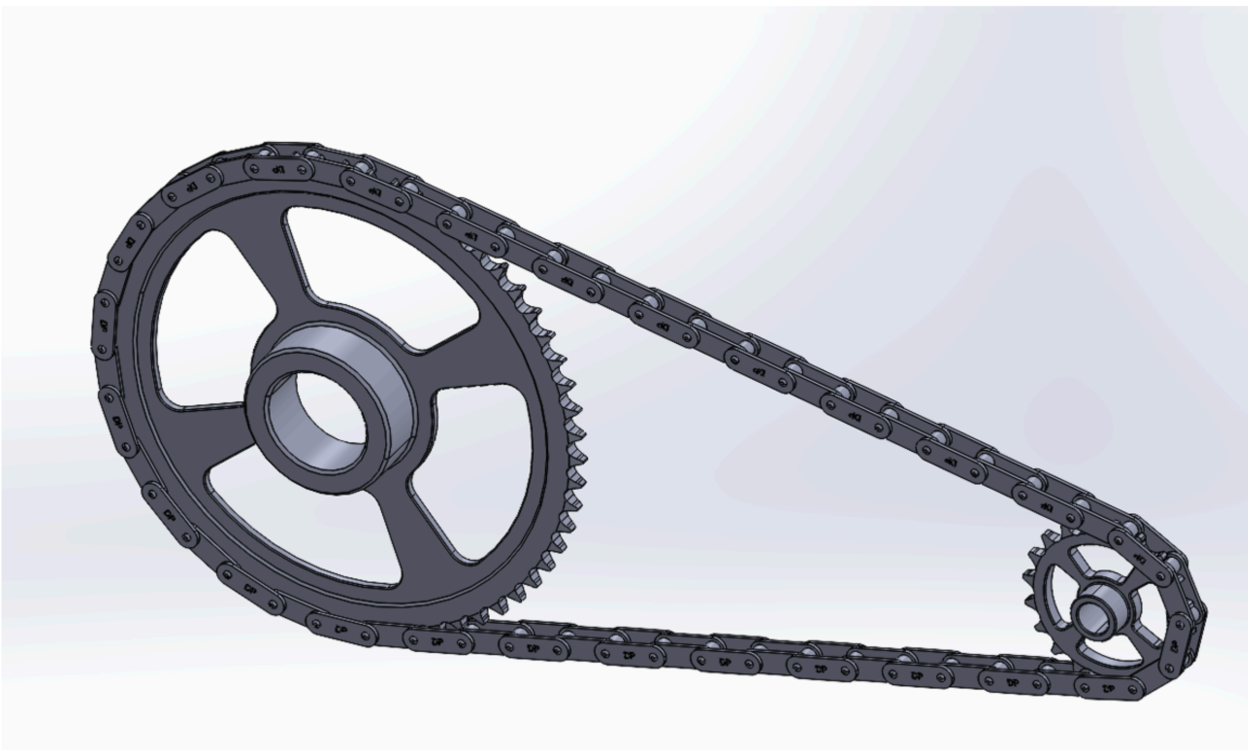


Figure 11: Model of Chain and Sprocket System on Solid Works

Table 5: Comparison of different materials for sprockets [17]

Property / Material	Stainless Steel	Carbon Steel	Aluminum	Plastic (Nylon/Acetal)	Cast Iron
Strength	High	Very High	Moderate	Low to Moderate	High

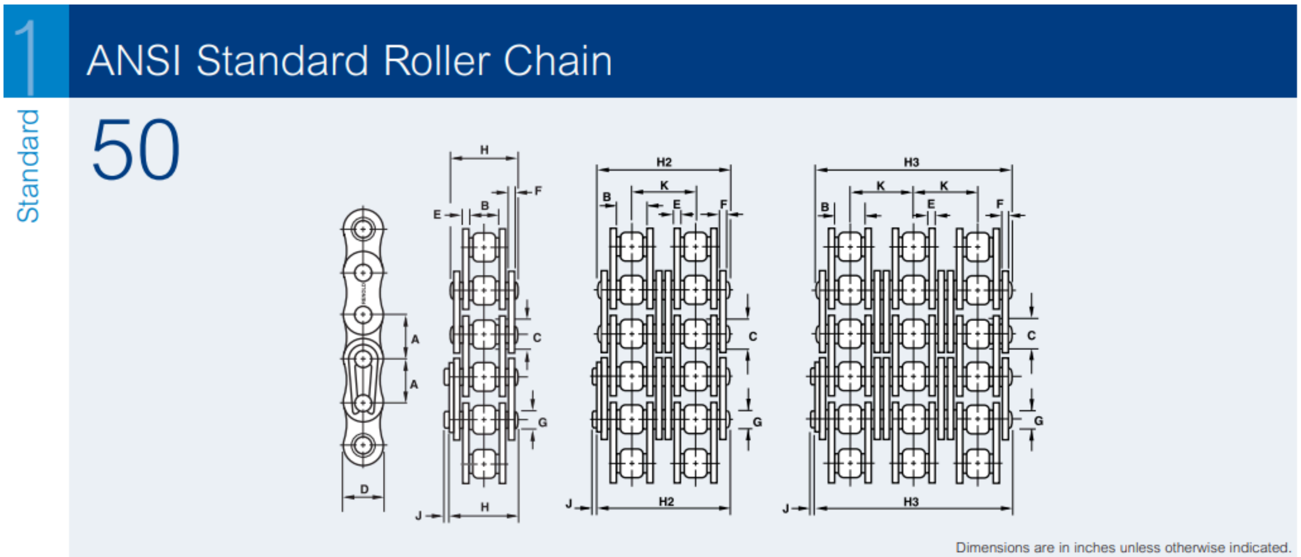
Corrosion Resistance	Excellent (rust-proof)	Poor (needs coating)	Good (forms oxide layer)	Excellent	Poor
Wear Resistance	Very Good	Excellent	Fair	Low	Good
Weight	High	High	Low (⅓ of steel)	Very Low	Very High
Machinability	Moderate	Good	Excellent	Excellent (moulded)	Fair
Cost	Moderate	Moderate	Low	Cheap	Low
Noise/Vibration	Moderate	Moderate to High	Low	Very Low	High
Thermal Stability	Excellent	Excellent	Good	Poor	Excellent
Applications	Heavy Duty outdoors	Heavy Duty indoors	Lightweight, low load	Conveyors, quiet systems	Heavy static loads

Chain Selection:
Since we have selected the Chain no. 50 below are the parameters from the catalog[15]. Fig 12 shows the same.

Table 5.1: Parameters of the chain

Parameter	Value
Chain No.	50
No. of Strands	1
Pitch	0.625 in
Inside Width	0.370 in
Minimum Roller Diameter	0.400 in
Maximum Plate Height	0.594 in
Maximum Inner Plate Thickness	0.080 in
Maximum Outer Plate Thickness	0.080 in
Maximum Pin Diameter	0.200 in
Maximum Pin Length	0.803 in

Connecting Link Extra	0.043 in
Minimum Tensile Strength	4,880 lbs
Rated Working Load	1,400 lbs
Weight	0.67 lbs/ft



Chain No.	No. of Strands	Pitch	Inside Width Min	Roller Diam Max	Plate Height Max	Inner Plate Thick Max	Outer Plate Thick Max	Pin Diam Max	Pin Length Max	Conn Link Extra Max	Trans Pitch	Tensile Strength Min	Rated Working Load	Weight
		A	B	C	D	E	F	G	H	J	K	Lbs	Lbs	Lbs/Ft
50	1	0.625	0.370	0.400	0.594	0.080	0.080	0.200	0.803	0.043	—	4,880	1,400	0.67

Fig 12: Chain parameters from the catalog

<u>Chain length calculation [18]</u>

Distance between the centres of the sprocket (CD) = 19.5625 inches
Number of teeth in large sprocket (Z_2) = 60
Number of teeth in small sprocket (Z_1) = 20
Chain pitch (p) = 0.625 inches

$$C = \frac{CD}{p} = 31.3 \quad (11)$$

$$S = Z_2 - Z_1 = 40 \quad (12)$$

$$A = Z_2 + Z_1 = 80 \quad (13)$$

S	T	S	T	S	T
1	0.03	35	31.03	69	120.60
2	0.10	36	32.83	70	124.12
3	0.23	37	34.68	71	127.69
4	0.41	38	36.58	72	131.31
5	0.63	39	38.53	73	134.99
6	0.91	40	40.53	74	138.71
7	1.24	41	42.58	75	142.48
8	1.62	42	44.68	76	146.31
9	2.05	43	46.84	77	150.18
10	2.53	44	49.04	78	154.11
11	3.06	45	51.29	79	158.09
12	3.65	46	53.60	80	162.11
13	4.28	47	55.95	81	166.19
14	4.96	48	58.36	82	170.32
15	5.70	49	60.82	83	174.50
16	6.48	50	63.33	84	178.73
17	7.32	51	65.88	85	183.01
18	8.21	52	68.49	86	187.34
19	9.14	53	71.15	87	191.73
20	10.13	54	73.86	88	196.16
21	11.17	55	76.62	89	200.64
22	12.26	56	79.44	90	205.18
23	13.40	57	82.30	91	209.76
24	14.59	58	85.21	92	214.40
25	15.83	59	88.17	93	219.08
26	17.12	60	91.19	94	223.82
27	18.47	61	94.25	95	228.61
28	19.86	62	97.37	96	233.44
29	21.30	63	100.54	97	238.33
30	22.80	64	103.75	98	243.27
31	24.34	65	107.02	99	248.26
32	25.94	66	110.34	100	253.30
33	27.58	67	113.71		
34	29.28	68	117.13		

Figure 13: Correlations for chain length calculation [18]

Now, we select T corresponding to S = 40 which is T = 40.53, to get chain length in pitches,

$$CL = 2C + \frac{A}{2} + \frac{T}{C} = 103.89 \tag{14}$$

Rounding off, we get the number of chain links to be 104, which, if multiplied by the pitch of 0.625 inches, we get chain length of 65 inches.

Subsection 4.5: Control algorithm for sun tracking

We are using the following equations of solar geometry to calculate the sunrise and sunset hours in order to adjust the microcontroller to start the motor at the standard sunrise clock time of the day and stop at the standard sunset clock time, and reset the position of the collector.[19][20]

For calculating the sunrise hour angle (ω_s),

$$Cos\,(\omega_s) = -\,tan\,(\Phi) * tan\,(\delta) \tag{15}$$

$$\omega_s = (12 - LAT) * 15 \tag{16}$$

Local apparent time (LAT) = Standard sunrise clock time – 4 × (Standard time longitude − Longitude of location) + (Equation of time correction) (17)

With the equation of time correction,

$$E = 229.18 * [0.000075 + 0.001868Cos(B) - 0.032077Sin(B) - 0.014615Cos(2B) - 0.04089Sin(2B)] \tag{18}$$

Where,

$$B = (n - 1) * \frac{360}{365} \tag{19}$$

Declination angle,

$$\delta = 23.45 * Sin\,[\frac{(284+n)*360}{365}] \tag{20}$$

n = Day of the year

Φ = Latitude of the location
We can calculate the standard sunrise time using this algorithm while getting day and location coordinates as input.
To calculate the sunset hours, we need to calculate the day length.

$$Day\ length\ = \frac{2}{15} * Cos^{-1}[-\ tan\ (\Phi)\ *\ tan\ (\delta)] \quad (21)$$

Standard sunset clock time = Standard sunrise clock time + Day length (22)

Thus we get the time of day and sunset time; we can get the sunrise time. This value of time is used to configure the system to initialize motion.

Subsection 4.6: Control algorithm for motor

For the motor control algorithm, we found out the gear ratio required and steps per second of the motor. Now ,we can use the Stepper Motor Controller SMC66 with housing, fulfilling our requirement.

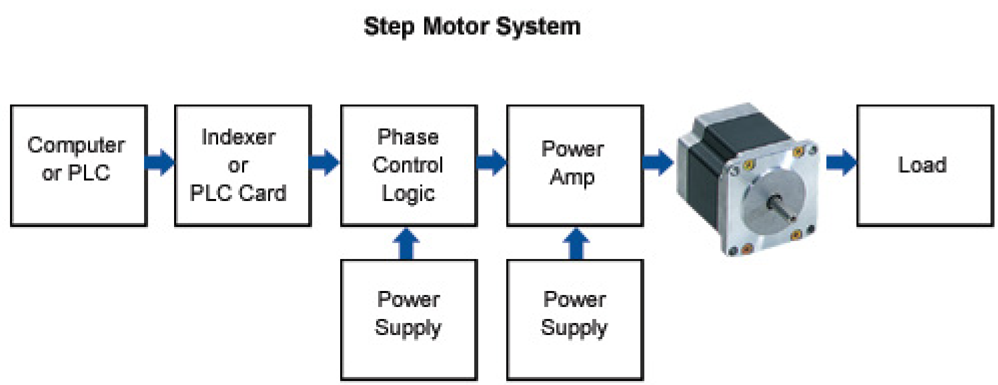


Figure 14: Motor control [21]

A typical stepper motor system consists of several essential components, each playing a key role in the motor's operation:

Computer or PLC: This serves as the control center for the entire system. It issues commands to control the stepper motor. The controller can range from a sophisticated PC/PLC to a simple push-button interface. [21]
Indexer or PLC Card: This component interprets commands from the computer and outputs pulses to drive the stepper motor. It adjusts the number and frequency of pulses to control motor acceleration, speed, and deceleration. Devices like the ORIENTAL MOTOR SG8030 indexer or pulse generator cards fall under this category. [21]
Phase Control Logic: This block determines the correct sequence for energizing the motor windings. It uses pulses from the indexer and energizes the phases accordingly. The phase control logic also includes a low-voltage power supply (typically around 5V) for the logic circuits. [21]
Power Supply and Amplifier: The motor power supply provides the necessary voltage to operate the motor (commonly 24V DC but may vary). The power amplifier acts as a switching device that energizes the motor phases in the correct order based on input from the phase control logic. [21]
Stepper Motor and Load: The motor executes the movement as dictated by the input pulses, transferring torque to the load, which is the PTC in our case. [21]

Together, these components ensure precise and controlled operation of the stepper motor system. For our application, we have a total of 50 steps in the 5 seconds of motion. Thus, 10 steps per second, so the time period of the pulses given by the PLC is 0.1 sec.

Subsection 4.7: Design Drawing and System Configuration

Fig. 14 is the system model designed in SolidWorks. It consists of the PTC, collector, receiver, the support system, and the chain-sprocket mechanism. Here, the motor model is not shown due to it's complex modelling, but in the system, there will be a motor fitted below, whose shaft goes through the driver sprocket. Also, the shafts of the other sprockets are connected to the support structure via ball bearings for power transmission, as shown in Fig. 14.

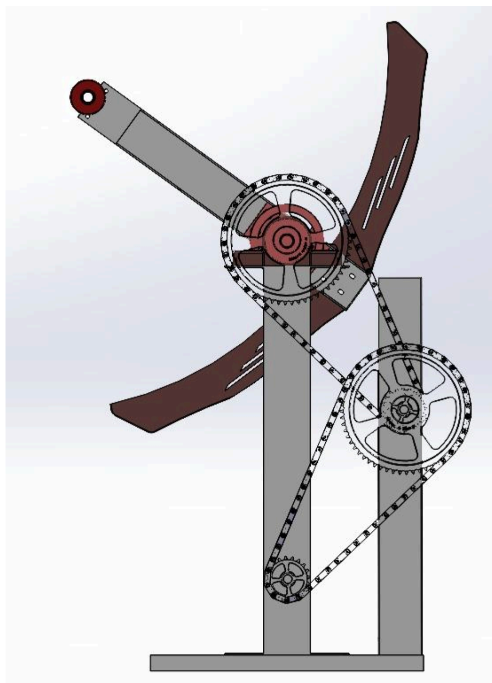


Figure 15: Solidworks model

Subsection 4.8: Material selection

Collector:

For the collector/reflector we choose Polymer-embedded silver on steel sheet[8], this is used because the

- **Steel Sheet** which acts as a base provides mechanical strength, rigidity, and durability. [22]
- And the **silver layer** has one of the highest reflectivities (low emissivity), which reduces infrared radiation loss and helps trap heat inside the absorber tube (as a selective surface).
- And the **polymer** is used to embed and protect the silver layer. [22]

Below table compares different standard materials used for the receiver on the PTC

Table 6: Comparison of different materials used for reflector [23]

Reflector Material	Reflectivity (%)	Durability (UV/Heat/Corrosion)	Cost

Polymer-embedded silver on steel sheet	92–96% (initial)	Moderate-High (depends on polymer stability)	Medium
Glass-mirrored reflectors (silver or aluminum on glass)	93–96%	Very high (if tempered glass is used)	High
Aluminum sheet with anodized or polished surface	85–90%	Moderate (can oxidize)	Low-Medium
Silvered polymer films (e.g., 3M Solar Mirror Film 1100/2000)	94–96%	Moderate	Medium
Aluminized polymer films (e.g., Mylar)	85–90%	Low-Moderate	Low

Receiver:
For the receiver, we chose stainless steel with optically selective coating because of its high strength, corrosion resistance, and ability to withstand high temperatures. When combined with an optically selective coating, it ensures high solar absorption and minimal thermal losses. This combination delivers durability, efficiency, and long-term performance in harsh outdoor environments. Below table compares different standard materials used for the receiver in the PTC.[24]

Table 7: Comparison of different standard materials used for the receiver [24]

Material System	**Solar Absorptance (α)**	**Thermal Emittance (ε)**	**Temp Tolerance**	**Durability**	**Cost**	**Remarks**
Stainless Steel + Selective Coating	0.94 – 0.96	0.05 – 0.10	~550°C	Excellent (corrosion & structural)	Moderate	Best balance of performance, durability, and cost
Black Chrome on Copper/Steel	0.90 – 0.95	0.10 – 0.15	~350°C	Moderate	Moderate	Older tech, decent performance but not ideal for CSP-scale
Cermet Coating + Glass Envelope	0.94 – 0.97	0.06 – 0.10	~500°C	Excellent (vacuum sealed)	High	Top performance; used in premium CSP systems

Painted or Anodized Aluminum	0.85 – 0.90	0.80 – 0.90	~200°C	Low (UV & heat degrade)	Low	Low-cost option for low-temp systems only

Section 5: Bill of Materials

Table 8: Costs of different components considered

Component	Material used	Quantity	Cost (INR)*
Chain (ANSI 50 Roller chain, pitch ⅝”)	Carbon steel	2	17,019.86 [25]
50B20SS Stainless Steel Sprocket	Stainless steel	2	76,617.02 [26]
50B60SS Stainless Steel Sprocket	Stainless steel	2	2,50,699.73 [27]
Stepper Motor	-	1	17,197.60 [28]
Planetary gearbox (1:20)	-	1	7,327.04 [29]
Ball bearing	Carbon chromium steel	2	17,605.10 [30]
Total cost			**3,86,466.35**

*Conversion rate on 14th March 2025

Inferences
1. The 50B60SS Sprockets (₹2.5 Lakh) dominate the cost — over 65% of the total.
2. The stepper motor + gearbox combo costs ₹24,524.64 (~6.3% of total), which is extremely cost-efficient for the torque/output it can deliver

Section 6: Other factors

Safety
- Riveted Chain #50 (0.625” pitch) offers non-detachable safety, ideal for high-speed/load applications.
- Sprocket Alignment must be precise to prevent derailment or wear.
- 4mm Hardened Glass protects components—inspect regularly for cracks or damage.
- Stainless Steel Receiver with selective coating—keep clean to maintain thermal efficiency.
- System Pressure up to 16 bar—monitor gauges and ensure relief valves function properly.
- Heat Transfer Fluid: Use water or propylene glycol (max 40%), ensuring non-toxic handling in case of leaks.

Operation

Load & Speed Compatibility:

- Chain #50 typically supports:
    - Working load: ~872 kgf (varies slightly by manufacturer)
    - Max speed: ~1200–1500 ft/min depending on lubrication and load
- The sprocket combination (20T/60T) gives a 3:1 speed reduction, meaning:
    - Chain speed remains the same, but output torque increases 3x.
    - Good for machinery requiring torque amplification.

Life and Durability

1. Expected Lifespan:
- For a well-maintained #50 riveted chain: Service life can exceed 15,000–20,000 hours in clean/lubricated environments.
- The Absolicon T160 is designed to have a long service life, generally around 20-30 years with proper maintenance. This is due to the robust materials used, including the stainless steel receiver and hardened glass, which are resistant to environmental stressors.
2. Wear Factors:
- 20T sprocket causes faster wear due to sharper articulation angles.
- 60T sprocket offers longer chain life due to gentler articulation and smoother engagement.
- Chain stretch occurs over time (elongation from pin/bushing wear) → replace if elongation exceeds 2–3% of original length.
3. Durability Enhancers:
- Use of hardened pins and bushings improves wear resistance.
- Nickel-plated or stainless variants offer corrosion resistance (if the environment is corrosive).
- Proper tensioning and periodic inspection significantly extend chain life.

Conclusion:

This project successfully demonstrates the design of a geared motor sun-tracking system for Parabolic Trough Collectors, ensuring enhanced solar thermal energy capture. By integrating precise torque calculations, reliable materials, and a well-engineered chain-sprocket mechanism, the system offers robust performance and durability. The use of an open-loop control strategy further simplifies the design while maintaining effective solar alignment.

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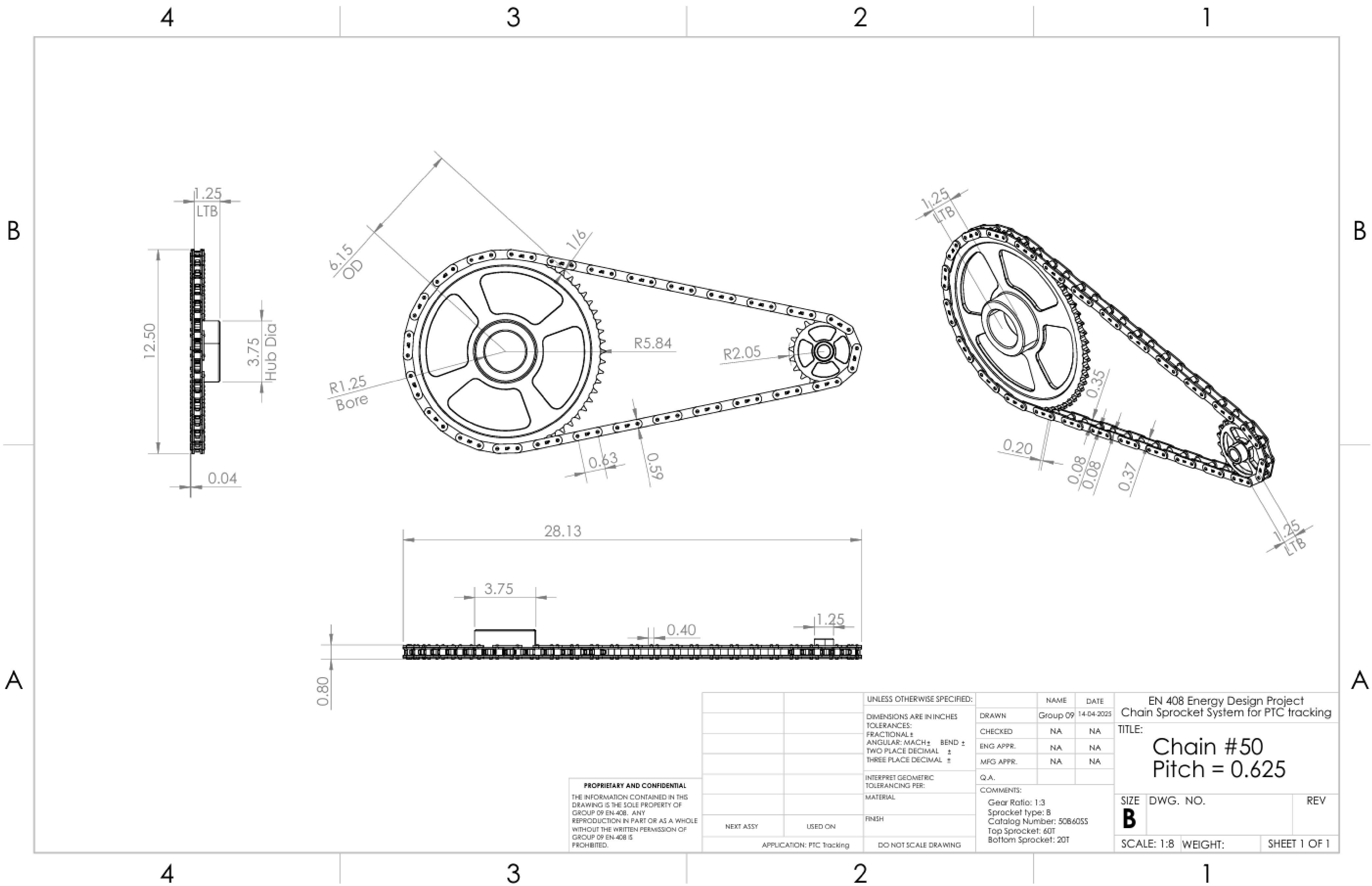

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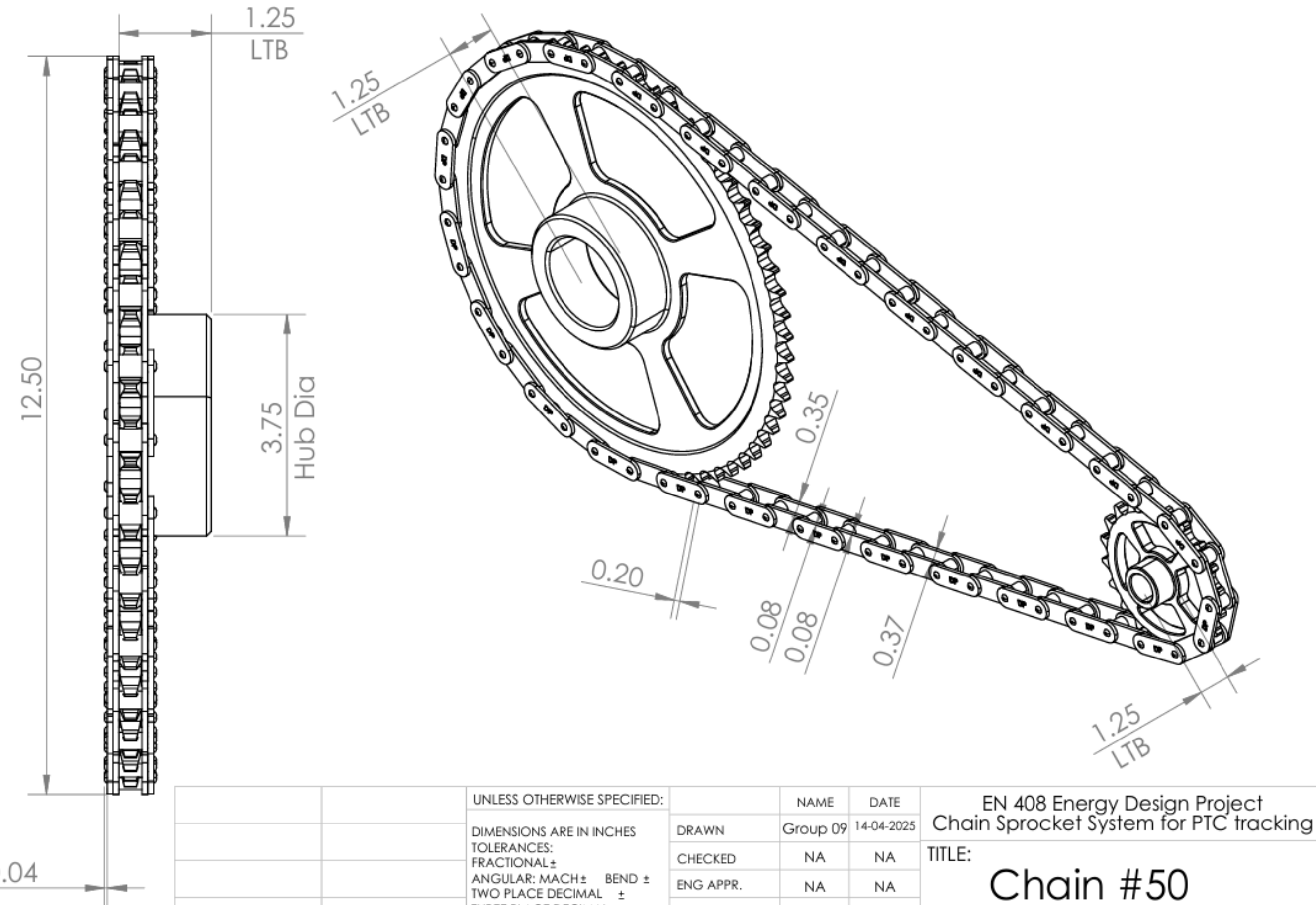

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Gear Ratio: 1:3
Sprocket type: B
Catalog Number: 50B60SS
Top Sprocket: 60T
Bottom Sprocket: 20T

EN 408 Energy Design Project
Chain Sprocket System for PTC tracking

TITLE:
Chain #50
Pitch = 0.625

SIZE **A** DWG. NO. REV

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